

Linear Algebra Techniques for Student Performance Analysis

1. Vectors and Matrices

Vectors represent one-dimensional data such as marks of one student. Matrices represent marks of many students across multiple subjects. Formula: $v=[x_1, x_2, \dots, x_n]$, $A=[a_{ij}]$.

VECTORS AND MATRICES

Vectors and matrices are fundamental tools in linear algebra and have wide applications in science, engineering, and technology.

1. VECTORS
A vector is an ordered list (column) of numbers.
Example: $v = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ (3-dimensional vector)

Basic Operations:
• Addition:
 $\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$
• Scalar Multiplication:
 $5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$

2. MATRICES
A matrix is a rectangular array of numbers arranged in rows and columns.
Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (2x2 matrix)

3. MATRIX-VECTOR MULTIPLICATION
If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $v = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$
Then $Av = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1(5) + 2(6) \\ 3(5) + 4(6) \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$

Matrices can be viewed as transformations that act on vectors.

5. IMPORTANT CONCEPTS
• Transpose: Swap rows and columns.
 $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
• Identity Matrix:
 $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. $AI = IA = A$
• Determinant (for 2x2 matrix):
If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
• Inverse Matrix:
If A^{-1} exists, $AA^{-1} = A^{-1}A = I$

KEY TAKEAWAYS
✓ Vectors are columns of numbers.
✓ Matrices are 2D arrays of numbers.
✓ Matrices can transform vectors.
✓ Matrix multiplication combines matrices.
✓ Transpose, determinant, inverse are important properties.

2. Dot Product, Cross Product, Projection and Norms

Dot product measures similarity. Cross product gives a perpendicular vector in 3D. Projection finds the component of one vector on another. Norm measures vector length.

1. DOT PRODUCT (SCALAR PRODUCT)
For two vectors:
 $A = (a_1, a_2, a_3)$, $B = (b_1, b_2, b_3)$
 $A \cdot B = a_1b_1 + a_2b_2 + a_3b_3$
Also:
 $A \cdot B = |A| |B| \cos \theta$
where:
 $|A|$ = magnitude of vector A
 $|B|$ = magnitude of vector B
 θ = angle between the vectors

2. CROSS PRODUCT (VECTOR PRODUCT)
For two vectors:
 $A = (a_1, a_2, a_3)$, $B = (b_1, b_2, b_3)$
 $A \times B = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$
Expanded form:
 $A \times B = (a_2b_3 - a_3b_2)i - (a_1b_3 - a_3b_1)j + (a_1b_2 - a_2b_1)k$
Magnitude:
 $|A \times B| = |A| |B| \sin \theta$

3. PROJECTION OF A VECTOR
Scalar Projection
Projection of vector A onto vector B :
 $\text{comp}_B(A) = \frac{A \cdot B}{|B|}$
Vector Projection
 $\text{proj}_B(A) = \left(\frac{A \cdot B}{B \cdot B} \right) B$
or equivalently,
 $\text{proj}_B(A) = \left(\frac{A \cdot B}{|B|^2} \right) B$

4. NORMS
1. L_1 Norm (Manhattan Norm)
 $\|x\|_1 = \sum_{i=1}^n |x_i|$
For $x = (x_1, x_2, \dots, x_n)$

2. L_2 Norm (Euclidean Norm)
 $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$
For 3D vector $x = (x, y, z)$
 $\|x\| = \sqrt{x^2 + y^2 + z^2}$

3. Infinity Norm (Maximum Norm)
 $\|x\|_\infty = \max(|x_1|, |x_2|, \dots, |x_n|)$
For $x = (x_1, x_2, \dots, x_n)$

4. General p-Norm
 $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$
where $p \geq 1$

3. LU and SVD Decomposition

LU decomposes $A=LU$. SVD decomposes $A=U\Sigma V^T$. These simplify solving systems, compression, and feature extraction.

LU DECOMPOSITION
LU decomposition factors a square matrix A into the product of a lower triangular matrix L and an upper triangular matrix U :
 $A = LU$

1. Definition
For a square matrix $A \in \mathbb{R}^{n \times n}$, LU decomposition is:
where
 $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$, $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

Note: Some definitions allow a permutation matrix P such that $PA = LU$ (Partial Pivoting) for numerical stability.

2. Properties
• L is unit lower triangular if diagonal entries of L are 1.
• U is upper triangular.
• If A is nonsingular, L and U are unique (with unit diagonal in L).
• Requires $\sim \frac{1}{2}n^2$ operations.

3. Example
Let $A = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ 0.25 & 0.5 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 4 & 3 & 2 \\ 0 & -0.5 & -1 \\ 0 & 0 & 2 \end{bmatrix}$

Check: $LU = A$

4. Applications
• Solving linear systems: $Ax = b$ (one forward and backward substitution).
• Matrix inversion.
• Determinant: $\det(A) = \det(L)\det(U) = \prod_{i=1}^n u_{ii}$. Since $\det(L) = 1$ if L is lower triangular.
• Used in numerical methods and optimization.

SVD DECOMPOSITION
Singular Value Decomposition factors any $m \times n$ matrix A into the product of three matrices: an orthogonal matrix U , a diagonal matrix of singular values Σ , and the transpose of an orthogonal matrix V^T :
 $A = U\Sigma V^T$

1. Definition
For any matrix $A \in \mathbb{R}^{m \times n}$,
where
 $U = \begin{bmatrix} u_1 & u_2 & \dots & u_m \\ u_{m+1} & u_{m+2} & \dots & u_m \\ \vdots & \vdots & \ddots & \vdots \\ u_{(m-1)n+1} & u_{(m-1)n+2} & \dots & u_{(m-1)n} \end{bmatrix}$ (is $m \times m$ orthogonal)
 $\Sigma = \begin{bmatrix} \sigma_1 & \sigma_2 & \dots & \sigma_r & 0 & \dots \\ 0 & 0 & \dots & 0 & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots \\ 0 & 0 & \dots & 0 & 0 & \dots \end{bmatrix}$ (is $m \times n$ diagonal)
 $V = \begin{bmatrix} v_1 & v_2 & \dots & v_n \\ v_{n+1} & v_{n+2} & \dots & v_n \\ \vdots & \vdots & \ddots & \vdots \\ v_{(n-1)m+1} & v_{(n-1)m+2} & \dots & v_{(n-1)m} \end{bmatrix}$ (is $n \times n$ orthogonal)
 $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$
 $r = \text{rank}(A)$

2. Properties
• Columns of U are orthonormal left singular vectors.
• Columns of V are orthonormal right singular vectors.
• Σ contains singular values $\sigma_1, \sigma_2, \dots, \sigma_r$ on the diagonal.
• Rows entries for any $m \times n$ matrix.
• Requires $\sim 4mn^2 \sim 8n^3/3$ operations (for $m \geq n$).

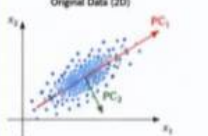
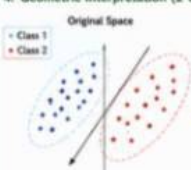
3. Applications
• Low-rank approximation (Truncated SVD).
• Solving least squares problems.
• Data compression and dimensionality reduction (PCA).
• Feature reduction in signal and image processing.
• Recommendation systems, machine learning.

Key Takeaway
LU is efficient for square systems and solving equations.
SVD is more general, gives insights, and is best for approximation and analysis.

Best Rank-k Approximation
 $A_k = U_k \Sigma_k V_k^T$
minimizes $\|A - B\|_F$ among all matrices B of rank k .

4. PCA and LDA

PCA reduces dimensions by preserving maximum variance. LDA reduces dimensions while maximizing separation between classes.

PCA (Principal Component Analysis)		LDA (Linear Discriminant Analysis)	
PCA is an unsupervised dimensionality reduction technique that transforms data into a new coordinate system so that the greatest variance comes to lie on the first coordinate (principal component), the second greatest variance on the second coordinate, and so on.		LDA is a supervised dimensionality reduction technique that finds a linear combination of features that best separates two or more classes.	
1. Goal Find directions (principal components) that maximize the variance of the data and project data onto these directions.	2. Method 1. Given data matrix $X \in \mathbb{R}^{n \times d}$ (n samples, d features) 2. Center the data: $X_c = X - 1\mu^T$, where $\mu = \frac{1}{n} \sum_{i=1}^n x_i$ 3. Compute covariance matrix: $\Sigma = \frac{1}{n-1} X_c^T X_c$ ($d \times d$ matrix) 4. Compute eigenvalues and eigenvectors of Σ : $\Sigma v_i = \lambda_i v_i$ 5. Sort eigenvectors by descending eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ 6. Form projection matrix $W_k = [v_1, v_2, \dots, v_k]$ (top k eigenvectors) 7. Project data: $Z = X_c W_k$ (reduced data in $\mathbb{R}^k$)	1. Goal Find projection directions that maximize class separability, i.e., maximize between-class scatter while minimizing within-class scatter.	2. Method 1. Given data $X \in \mathbb{R}^{n \times d}$ with class labels from C classes 2. Compute mean vectors: Overall mean $\mu = \frac{1}{n} \sum_{i=1}^n x_i$ Class mean $\mu_c = \frac{1}{n_c} \sum_{x_i \in c} x_i$ 3. Compute scatter matrices: Within-class scatter: $S_W = \sum_{c=1}^C \sum_{x_i \in c} (x_i - \mu_c)(x_i - \mu_c)^T$ Between-class scatter: $S_B = \sum_{c=1}^C n_c (\mu_c - \mu)(\mu_c - \mu)^T$ 4. Solve generalized eigenvalue problem: $S_B w = \lambda S_W w$ 5. Choose top $(C-1)$ eigenvectors corresponding to largest eigenvalues to form projection matrix W 6. Project data: $Z = XW$
3. Key Equations Covariance matrix: $\Sigma = \frac{1}{n-1} X_c^T X_c$ Eigen decomposition: $\Sigma v_i = \lambda_i v_i$ Projection: $Z = X_c W_k$	4. Geometric Interpretation  5. Properties - Unsupervised (no class labels used) - Captures maximum variance - Components are orthogonal - Sensitive to feature scaling	3. Key Equations Within-class scatter: $S_W = \sum_{c=1}^C \sum_{x_i \in c} (x_i - \mu_c)(x_i - \mu_c)^T$ Between-class scatter: $S_B = \sum_{c=1}^C n_c (\mu_c - \mu)(\mu_c - \mu)^T$ Objective (to maximize): $J(w) = \frac{w^T S_B w}{w^T S_W w}$	5. Properties - Supervised (uses class labels) - Finds directions that best separate classes - At most $(C-1)$ meaningful components for C classes - Assumes class distributions are Gaussian with equal covariance (for optimality)
6. Applications - Dimensionality reduction and visualization - Data compression - Noise reduction - Feature extraction - Preprocessing for machine learning	7. Explained Variance Proportion of variance explained by the i -th principal component: $\text{Explained Variance Ratio}_i = \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}$ Cumulative explained variance up to k components: $\sum_{i=1}^k \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}$	4. Geometric Interpretation (2 Classes)  6. Applications - Dimensionality reduction with class information - Classification - Feature extraction - Face recognition, bioinformatics, text classification, etc. 7. Notes - For 2 classes, LDA finds 1 projection direction. - For multi-class, LDA finds up to $(C-1)$ directions. - Computational cost is lower than PCA when $(C-1) \ll d$. - Sensitive to outliers and non-Gaussian distributions.	

5. Eigenvalues and Eigenvectors

Eigenvectors are important directions of a matrix. Eigenvalues indicate the amount of variance or importance along each eigenvector..

EIGENVALUES AND EIGENVECTORS

For a square matrix $A \in \mathbb{R}^{n \times n}$, a nonzero vector $v \in \mathbb{R}^n$ is an eigenvector of A if there exists a scalar $\lambda \in \mathbb{R}$ (an eigenvalue) such that

$$Av = \lambda v$$

The eigenvalue λ represents the factor by which A scales the eigenvector v (possibly changing its direction).

Geometric Interpretation (2D)

v is an eigenvector
If $\lambda > 0$: same direction
If $\lambda < 0$: opposite direction
If $0 < \lambda < 1$: shorter vector
If $\lambda > 1$: longer vector

1. HOW TO FIND EIGENVALUES

Solve the characteristic equation:

$$\det(A - \lambda I) = 0$$

where I is the $n \times n$ identity matrix.

The roots $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A .

2. HOW TO FIND EIGENVECTORS

For each eigenvalue λ , solve

$$(A - \lambda I)v = 0$$

The solutions $v \neq 0$ form the eigenspace corresponding to λ .

4. EXAMPLES

Example 1: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

Characteristic equation:

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = 0$$
$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0 \Rightarrow (\lambda-1)(\lambda-3) = 0$$

Eigenvalues: $\lambda_1 = 1, \lambda_2 = 3$

For $\lambda_1 = 1$: $(A - I)v = 0 \Rightarrow$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} v = 0 \Rightarrow v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For $\lambda_2 = 3$: $(A - 3I)v = 0 \Rightarrow$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} v = 0 \Rightarrow v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Example 2: $A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = (4-\lambda)(1-\lambda) = 0$$

Eigenvalues: $\lambda_1 = 4, \lambda_2 = 1$

For $\lambda_1 = 4$: $(A - 4I)v = 0 \Rightarrow$

$$\begin{bmatrix} 0 & 0 \\ 0 & -3 \end{bmatrix} v = 0 \Rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda_2 = 1$: $(A - I)v = 0 \Rightarrow$

$$\begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} v = 0 \Rightarrow v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

3. PROPERTIES

- Eigenvalues of a real matrix can be real or complex.
- Sum of eigenvalues = $\text{trace}(A)$: $\sum_{i=1}^n \lambda_i = \text{tr}(A)$
- Product of eigenvalues = $\det(A)$: $\prod_{i=1}^n \lambda_i = \det(A)$
- Eigenvectors corresponding to distinct eigenvalues are linearly independent.
- If A is symmetric ($A^T = A$):
 - All eigenvalues are real.
 - Eigenvectors corresponding to different eigenvalues are orthogonal.

5. SPECIAL CASES

Diagonal matrix

If $A = \text{diag}(a_1, a_2, \dots, a_n)$, then

Eigenvalues: $\lambda_i = a_i$

Eigenvectors: standard basis vectors e_i

Triangular matrix

Eigenvalues are the diagonal entries of A .

Symmetric matrix ($A^T = A$)

Has n real eigenvalues and an orthogonal basis of eigenvectors.

Orthogonality (Symmetric A)

If A is symmetric and

$Av_1 = \lambda_1 v_1, Av_2 = \lambda_2 v_2$

with $\lambda_1 \neq \lambda_2$, then

$$v_1^T v_2 = 0$$

(eigenvectors are orthogonal)

6. SUMMARY TABLE

Concept	Equation / Condition	Meaning
Eigenvalue	$\det(A - \lambda I) = 0$	Values of λ for which non-trivial solutions exist.
Eigenvector	$(A - \lambda I)v = 0, v \neq 0$	Direction that only gets scaled by A .
Trace relation	$\sum_{i=1}^n \lambda_i = \text{tr}(A)$	Sum of eigenvalues equals the trace.
Determinant relation	$\prod_{i=1}^n \lambda_i = \det(A)$	Product of eigenvalues equals the determinant.
Independence	$\lambda_i \neq \lambda_j \Rightarrow v_i \perp v_j$ (if A symmetric)	Eigenvectors of distinct eigenvalues are orthogonal (symmetric case).
Diagonalization	$A = PDP^{-1}$	If A has n linearly independent eigenvectors; D is diagonal of eigenvalues.

7. APPLICATIONS

- Principal Component Analysis (PCA)
- Stability analysis in differential equations and dynamical systems
- Vibrations and structural engineering
- Markov chains and stochastic processes
- Image compression and signal processing
- Quantum mechanics and Google PageRank